BUILDING NAT AND DHCP TOGETHER

STEP 1:

ADDING DHCP CONFIGURATION TO INTERFACE GIGA0/0/0:

NETWORK-1(HOME):

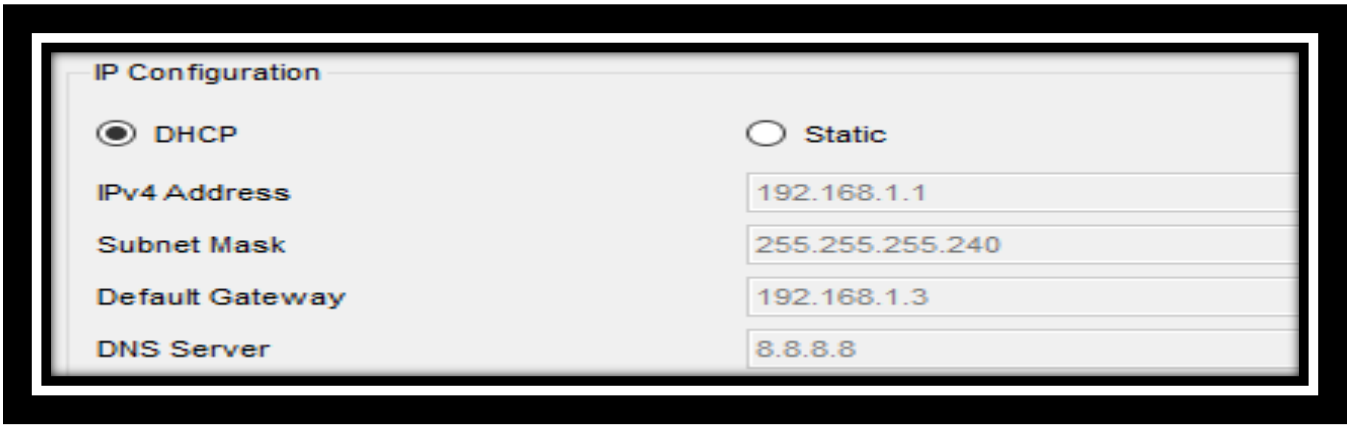
```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/0
Router(config-if)#ip add 192.168.1.3 255.255.255.240
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.3 192.168.1.6
Router(config)#ip dhcp pool home
Router(dhcp-config)#network 192.168.1.0 255.255.255.240
Router(dhcp-config)#default-router 192.168.1.3
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
```

OUTPUT:



STEP 2:

ADDING DHCP CONFIGURATION TO INTERFACE GIGA0/0/1:

NETWORK-1(OFFICE):

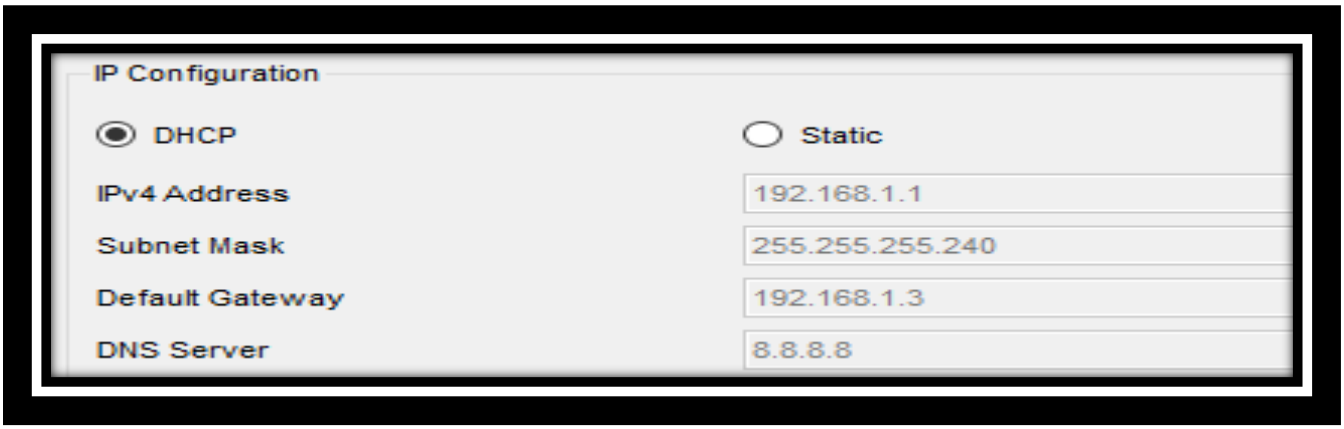
```
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.3 192.168.1.6
Router(config)#ip dhcp pool home
Router(dhcp-config)#network 192.168.1.0 255.255.255.240
Router(dhcp-config)#default-router 192.168.1.3
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#interface giga0/0/1
Router(config-if)#ip add 192.168.2.3 255.255.255.240
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.2.3 192.168.2.7
Router(config)#ip dhcp pool office
Router(dhcp-config)#network 192.168.2.0 255.255.255.240
Router(dhcp-config)#default-router 192.168.2.3
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#
```

OUTPUT:



VERFIED OUTPUT FOR DHCP PING

```
Reply from 192.168.2.8: bytes=32 time<1ms TTL=127
Reply from 192.168.2.8: bytes=32 time<1ms TTL=127
Reply from 192.168.2.8: bytes=32 time=1ms TTL=127
Reply from 192.168.2.8: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

NAT IMPLEMENTING ROUTER TO SERVER

(NETWORK ADDRESS TRANSLATION)

STEP 1:

ROUTER CONFIGURATION :

ASSIGN IP ADDRESS THEN DEPLOYING NAT :

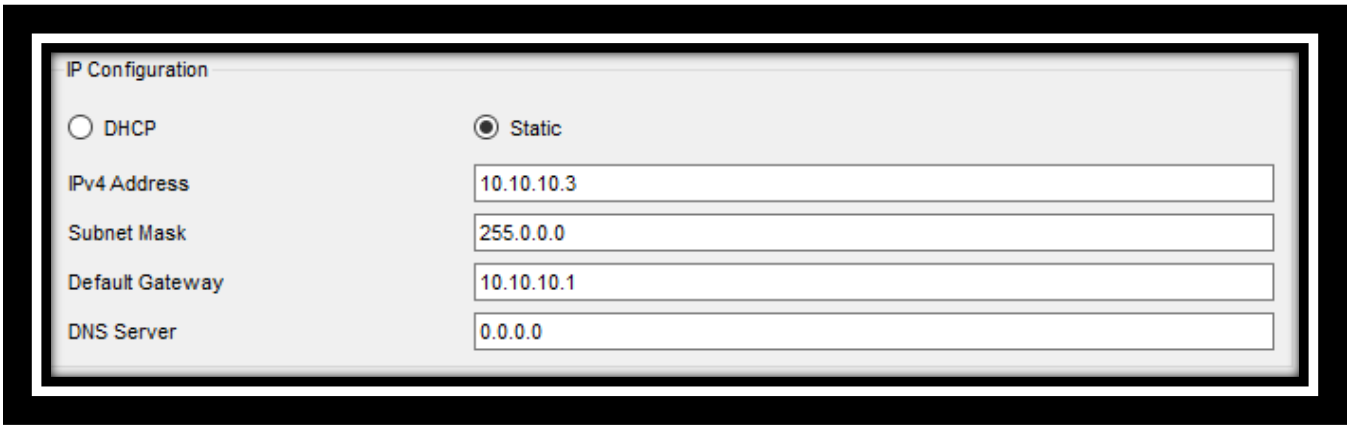
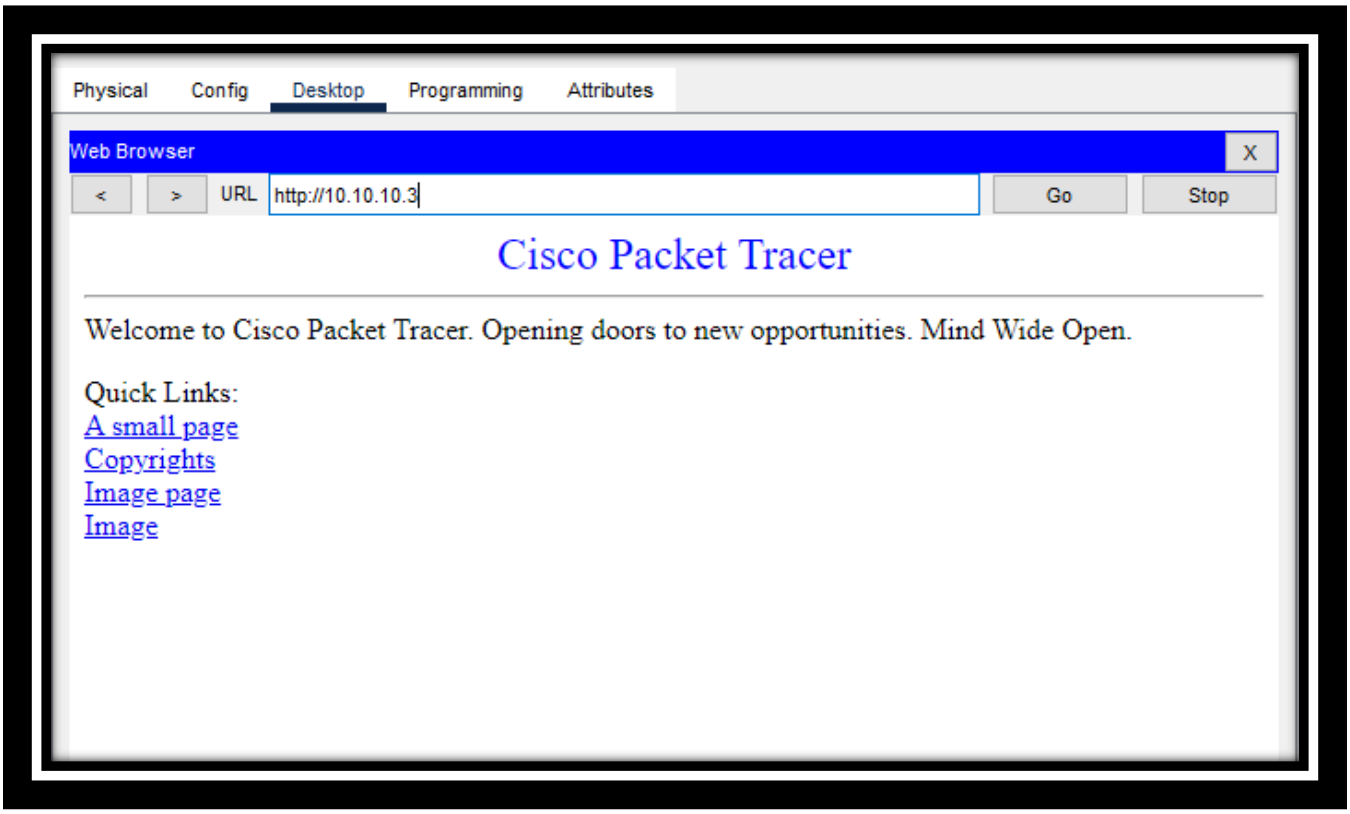
```
Router>
Router>
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/2
Router(config-if)#ip add 10.10.10.1 255.255.255.248
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/2, changed state to up

Router(config-if)#exit
Router(config)#interface giga0/0/1
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface giga0/0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#
```

ASSIGN IP ADDRESS TO SERVER:



FINALL VERIFICATION FOR NAT WORKING